

SPM MRI Course
Advanced fMRI

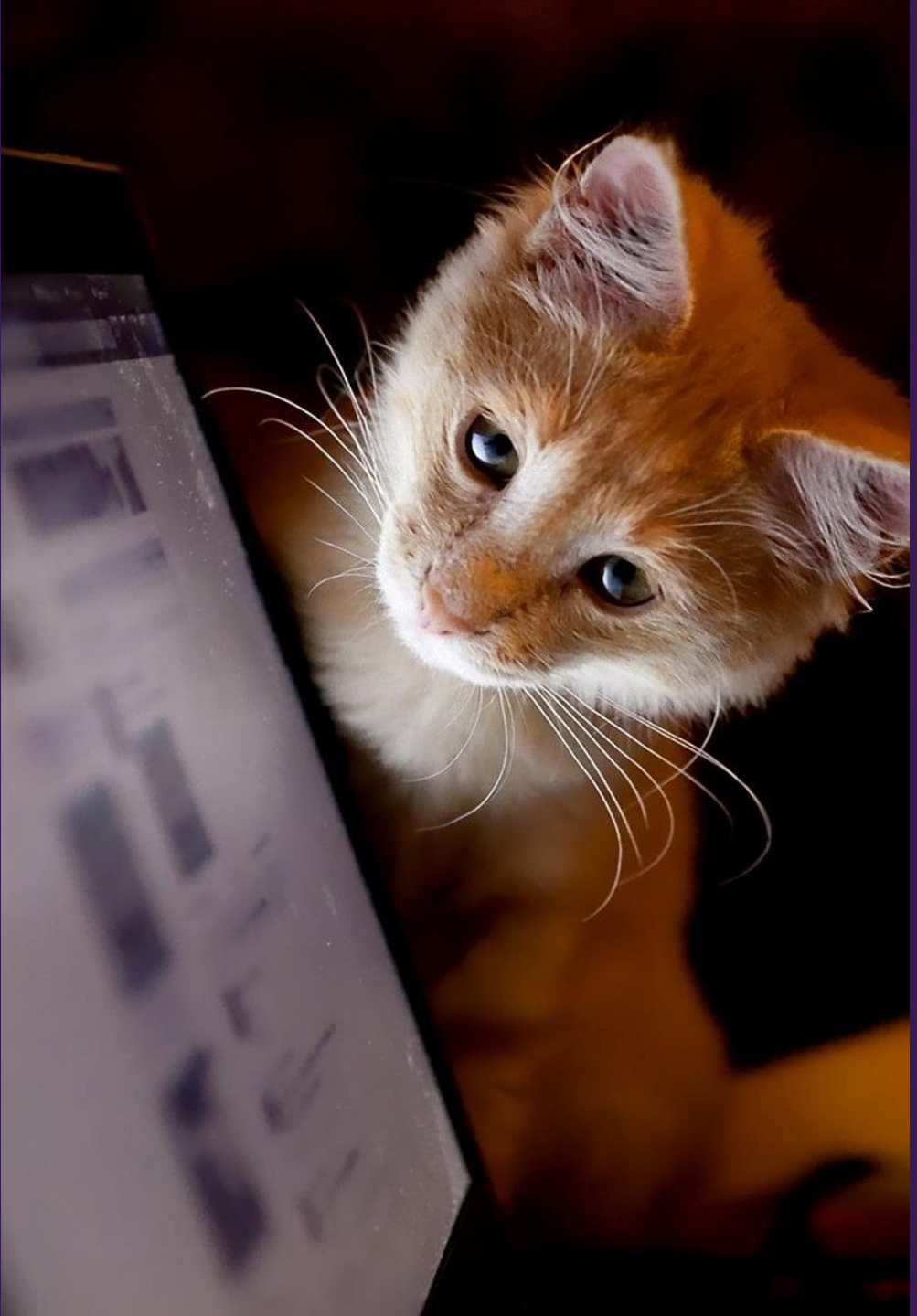
19-21 October 2026

Functional Imaging Laboratory
Department of Imaging Neuroscience



Overview of the session

- Visual **quality control** of the data
- **Motion**-related problems and how to fix them
- Accounting for **variability** – within and between individuals
- Optimising experimental **design**
- Bonus content – **manipulating images**

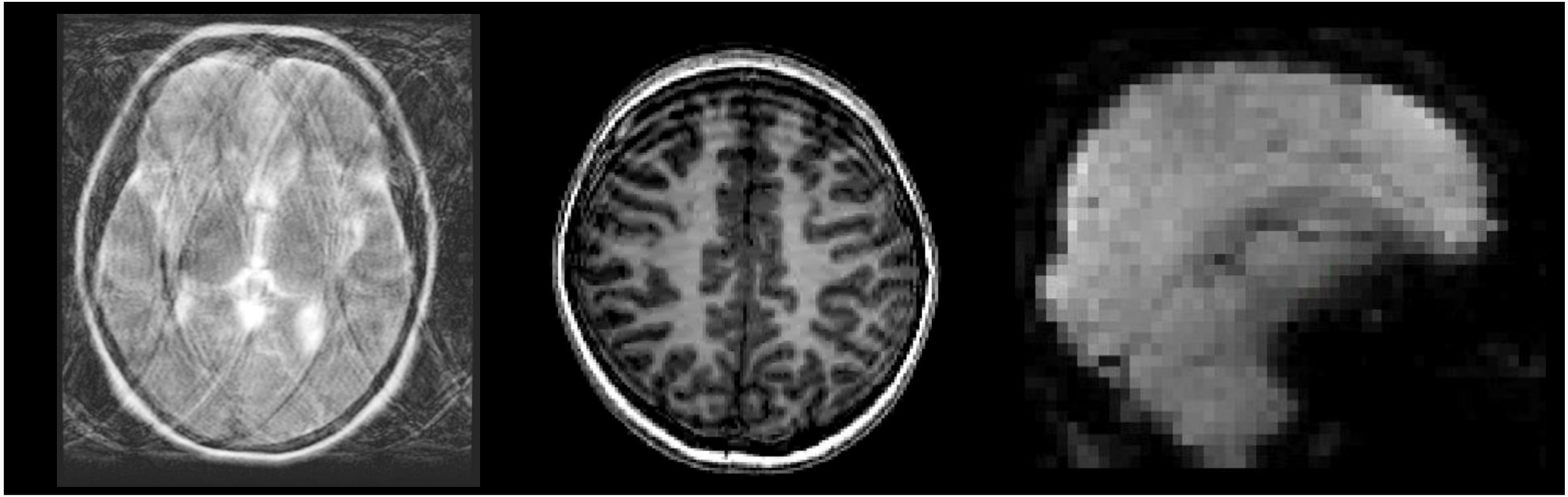


Visual quality control

AKA, always look at your data!

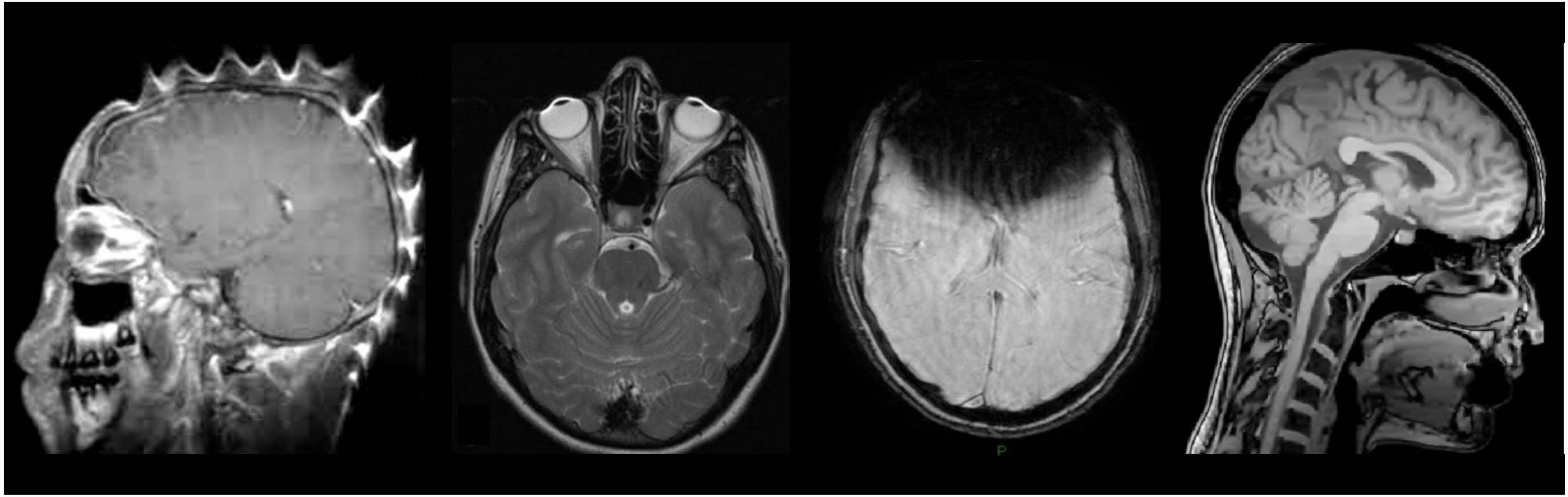
Visual inspection

Motion



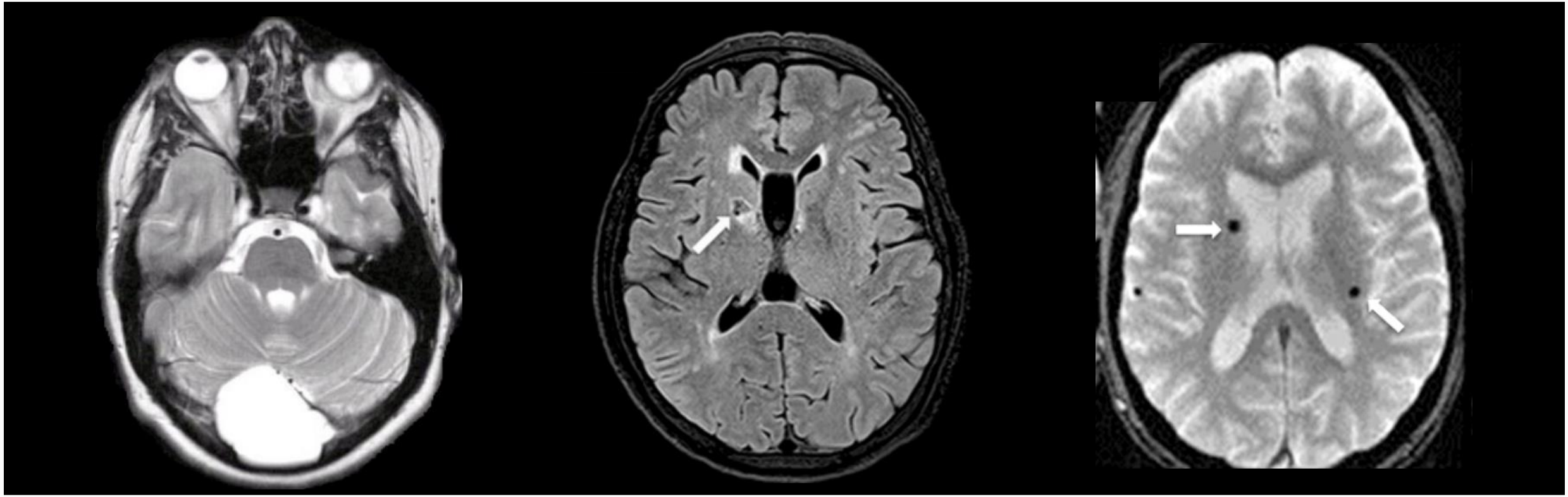
Visual inspection

Susceptibility artefacts



Visual inspection

Incidental findings



Visual inspection

A how-to



or



Practical examples here: link.

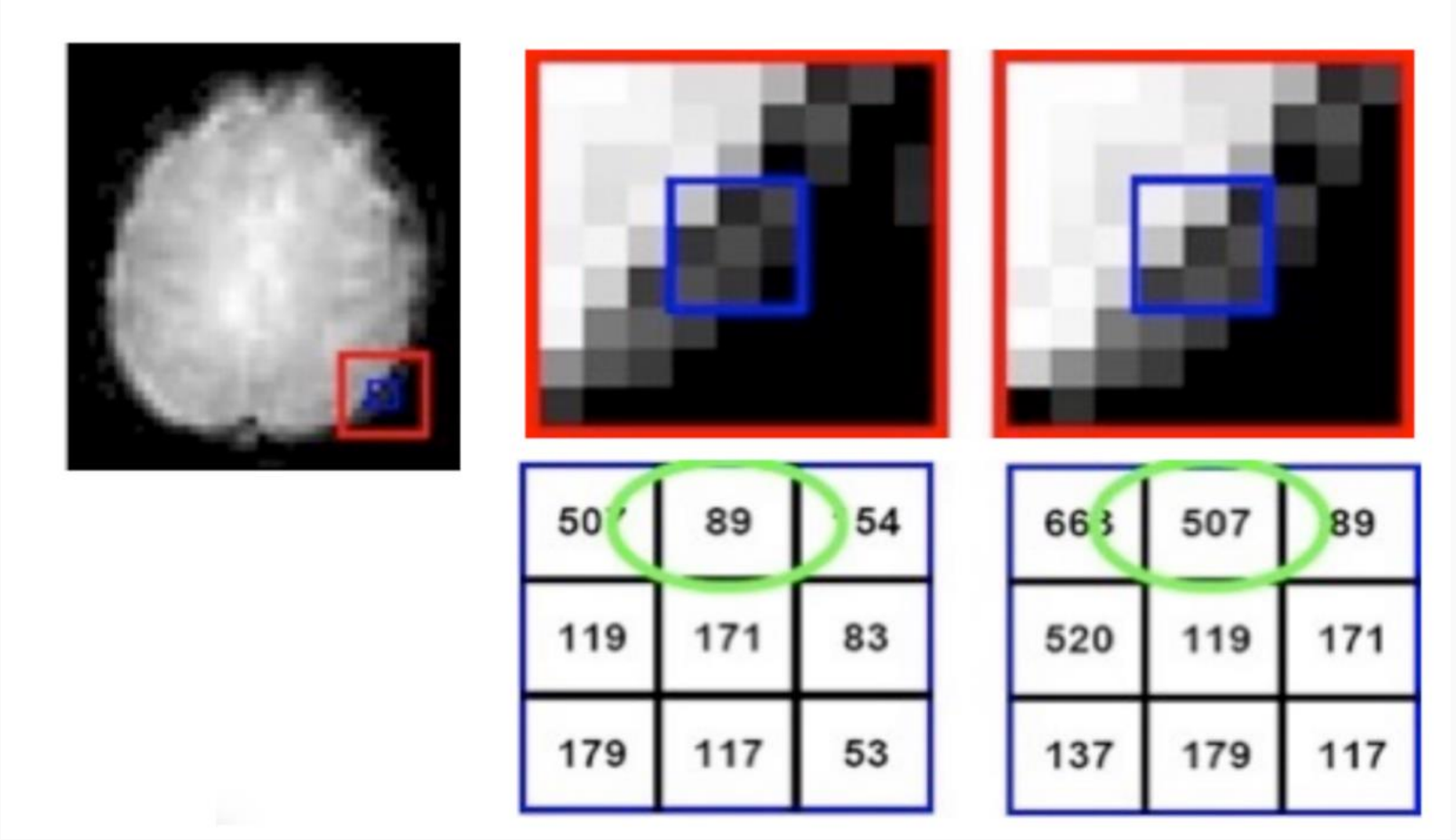


Advanced motion correction

Fixing one of the biggest confounds in your data

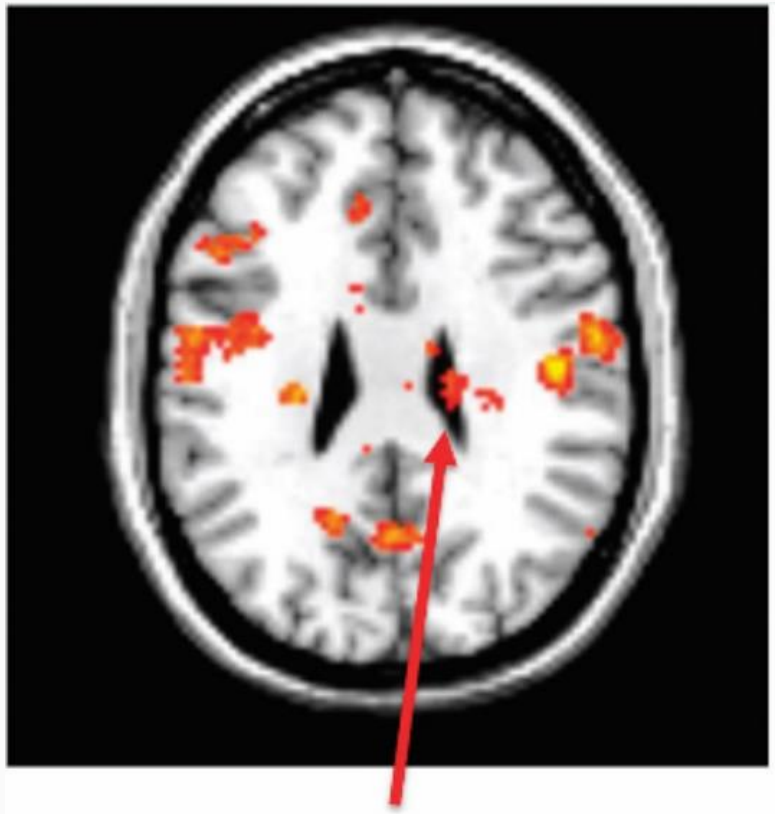
Why is motion a problem?

Spurious activations

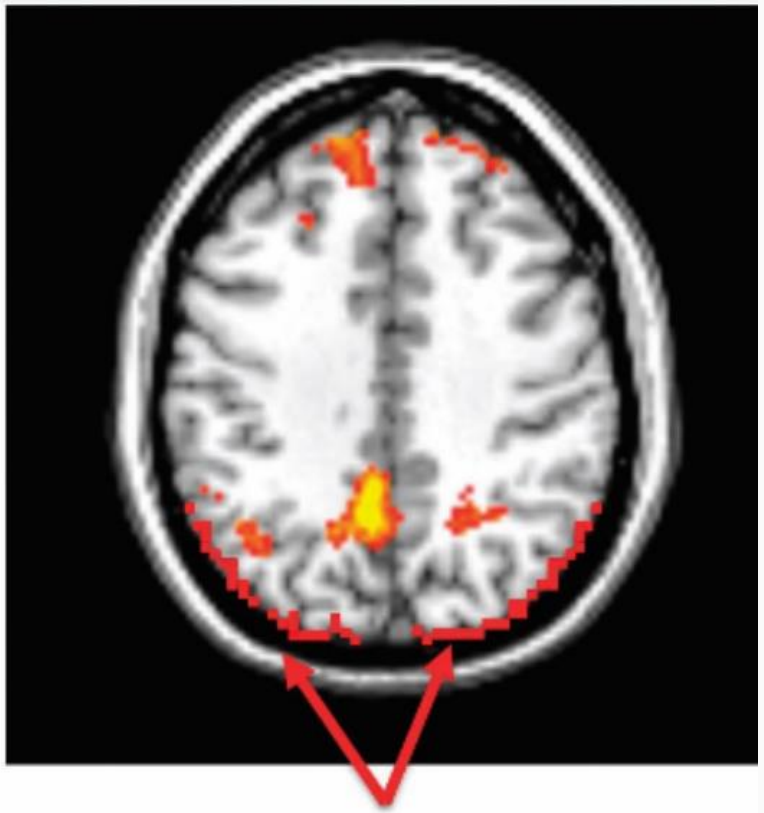


Why is motion a problem?

Spurious activations



CSF artefact



Ring artefact

Approaches to dealing with motion

A how-to

1. **Quantify** motion and **exclude** participants based on a priori thresholds.
2. **Censor** the **noisiest volumes** in the scan.
3. Model **additional** motion **parameters**.

...other approaches are also available.

Practical examples in
`motion.mlx`

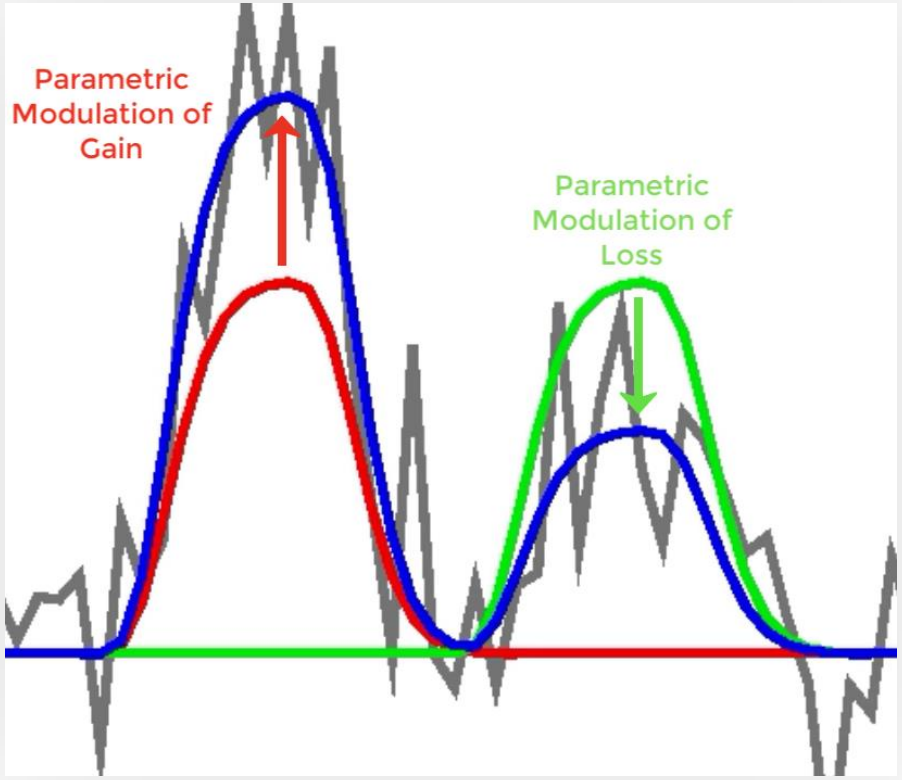


Accounting for variability

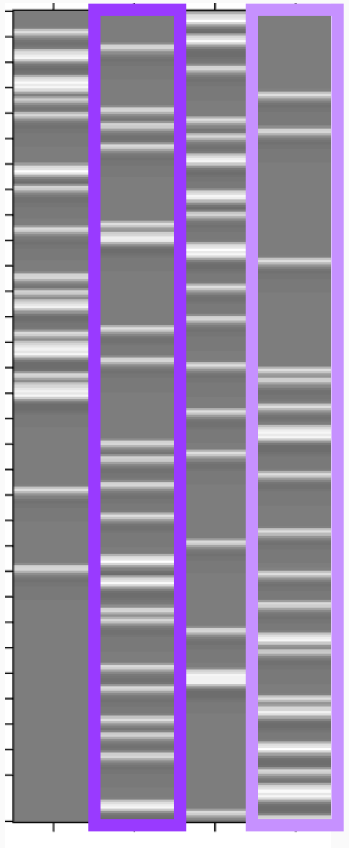
Additional regressors for subject and group-level analyses

Accounting for within-participant variability

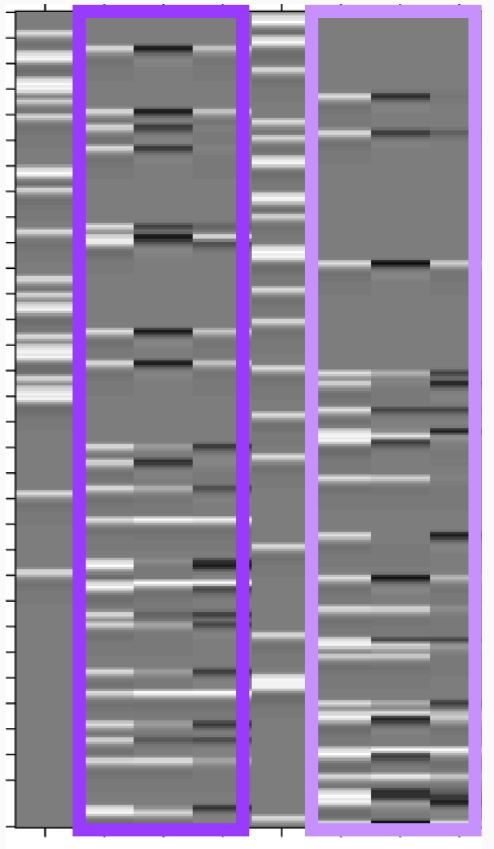
Parametric modulation



Regular GLM



Parametric modulation GLM

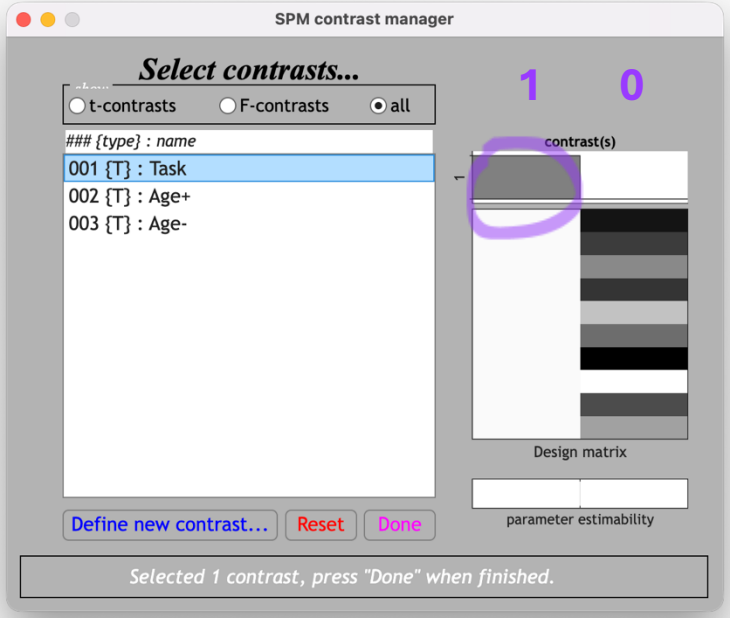


Practical examples in the **parametric_modulation** directory and here: link.

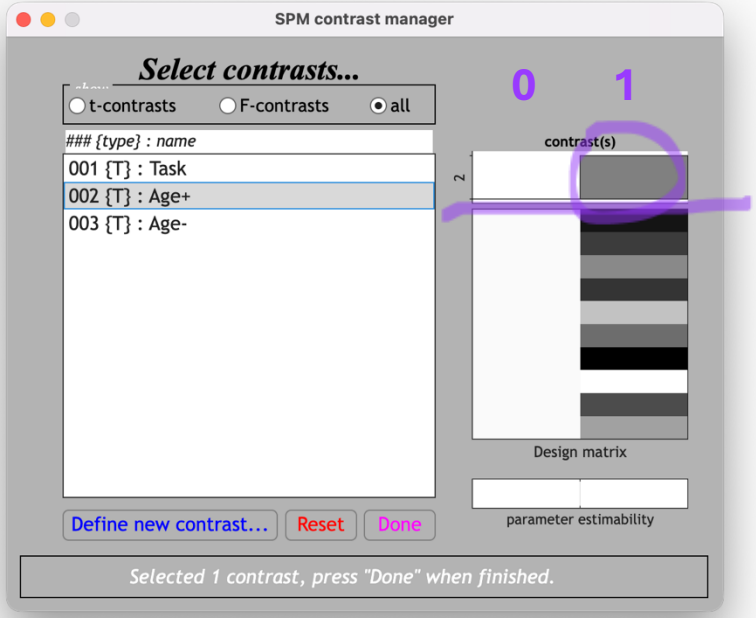
Accounting for between-participant variability

Group-level regressors

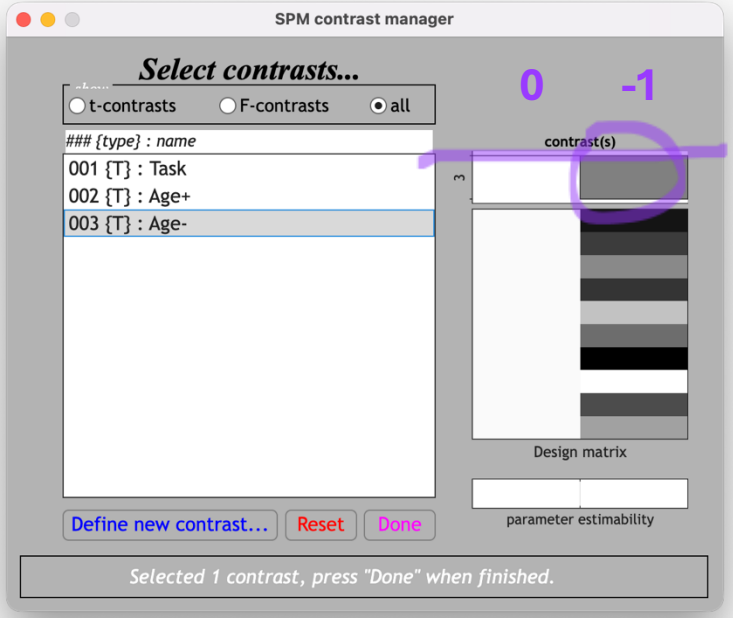
Do my task effects remain if I covary for age?



Are my task effects **positively** associated with age?



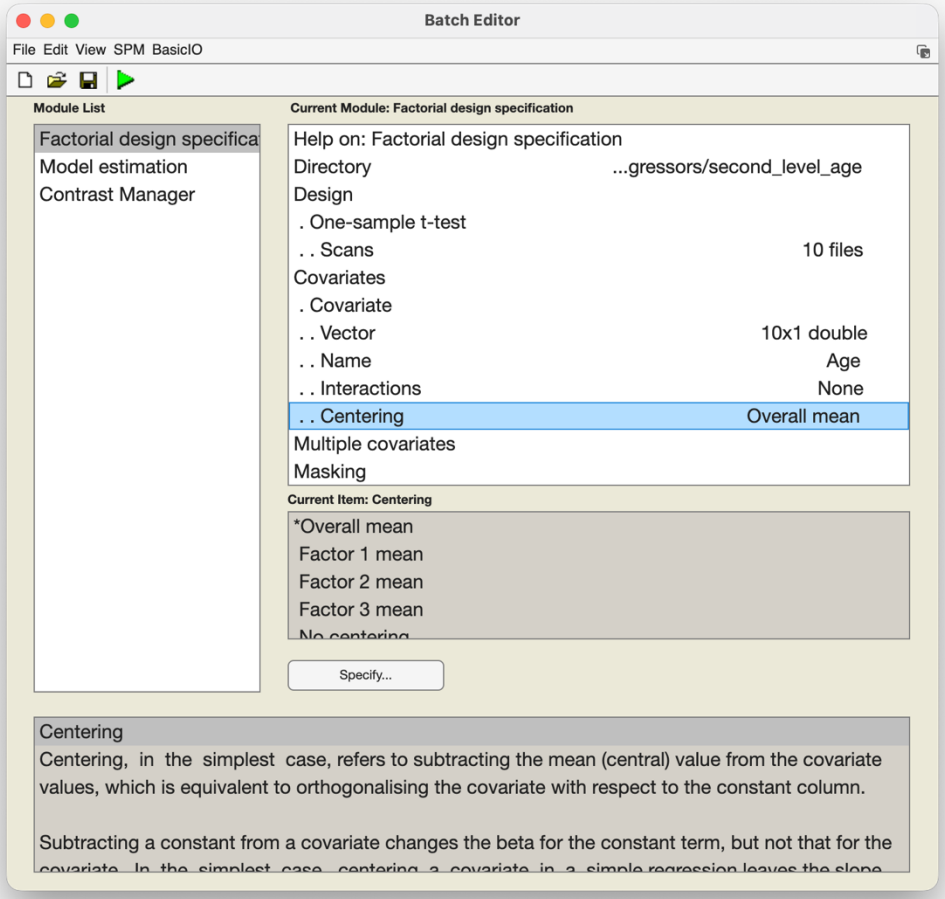
Are my task effects **negatively** associated with age?



Practical examples in the **group_regressors** directory.ç

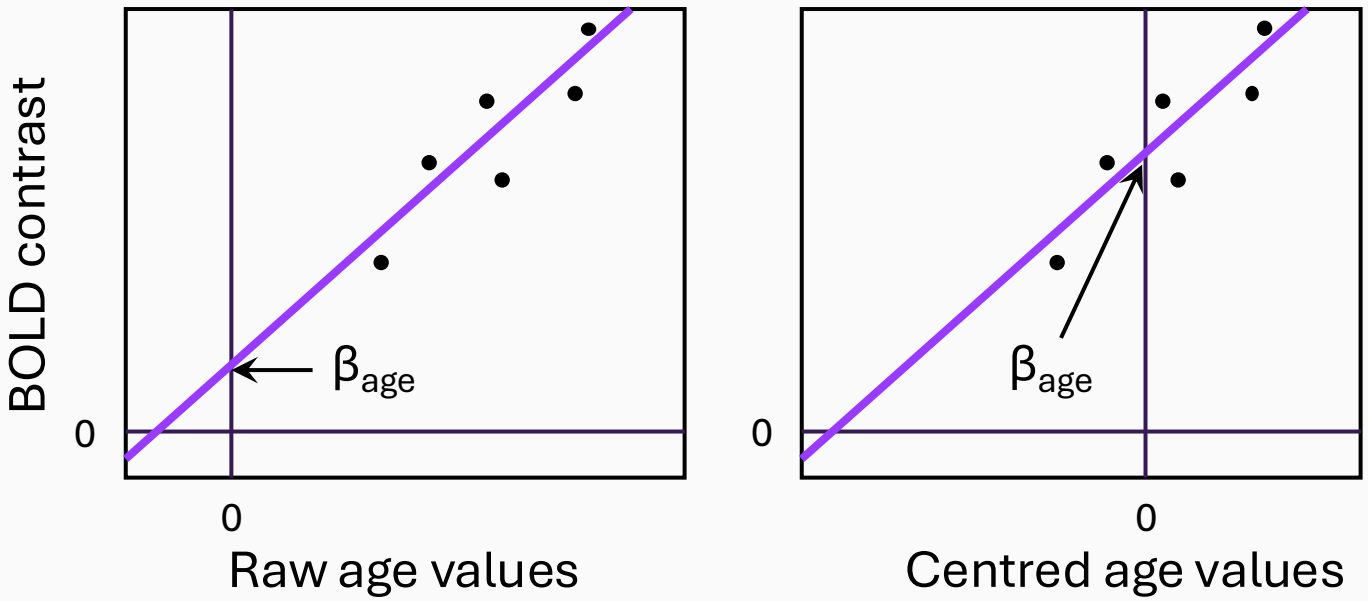
Accounting for between-participant variability

Group-level regressors

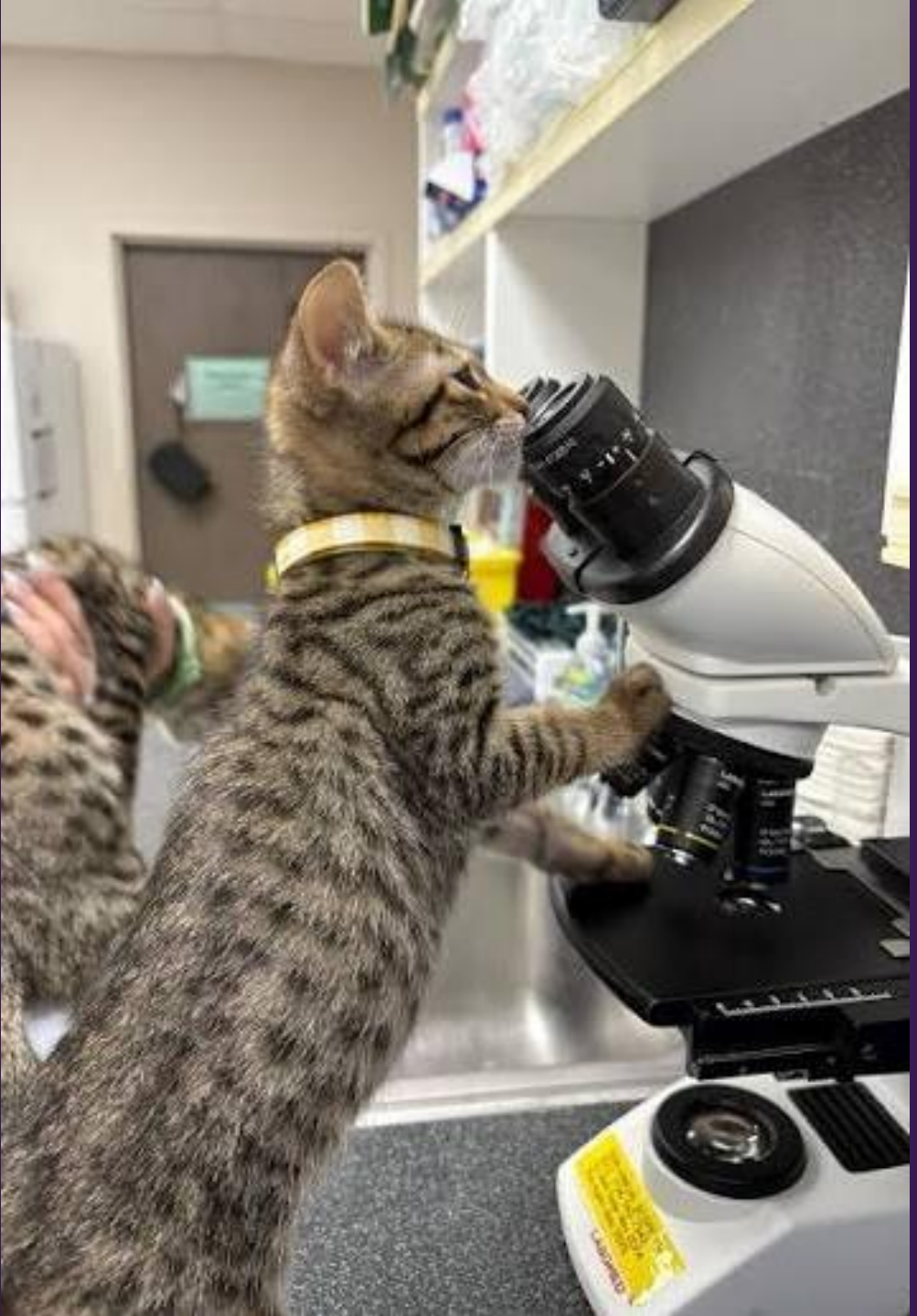


Mean centring regressors changes their interpretation

β_{age} always represents BOLD at age = 0



Great resource here: link.



Optimizing experimental design

Planning your tasks ahead of data collection

Theory [1/2] – the t-statistic

To maximize the t-statistic for an interesting effect, we need to minimize the covariance among the regressors.

Why? With contrast vector c , to test the null hypothesis:

$$H_0: c^T\beta = 0$$

The t-statistic is:

$$t = \frac{c^T\hat{\beta}}{\sqrt{\hat{\sigma}^2 c^T(X^TX)^{-1}c}}$$

The more covariance between regressors (related to the contrast), the greater $c^T(X^TX)^{-1}c$, and the smaller the t-statistic

Tip: The efficiency of a contrast C can be quantified by: $\text{efficiency(C)} = trace[C^T(X^TX)^{-1}C]^{-1}$

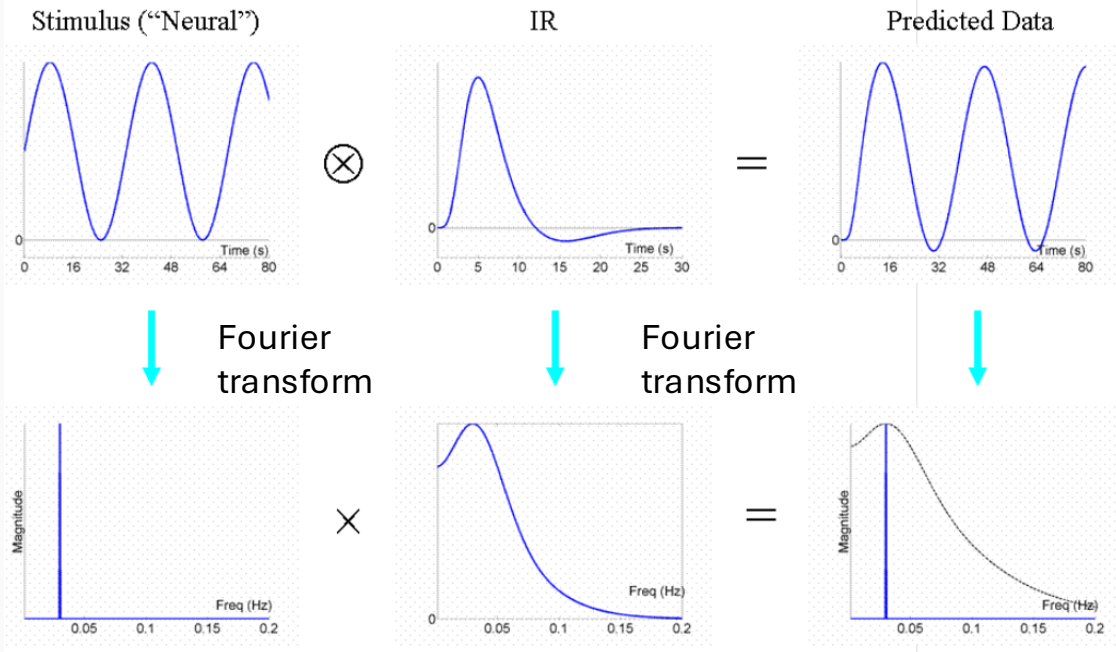
Theory [2/2] – stimulus timing for fMRI

We need to time our trials to be in the right frequency range for the haemodynamic response

Why? **Convolution** in the time domain (top row) equals **multiplication** in the frequency domain (bottom row). The Haemodynamic Response Function (HRF) is like a filter – preserving neural frequencies around 0.03Hz

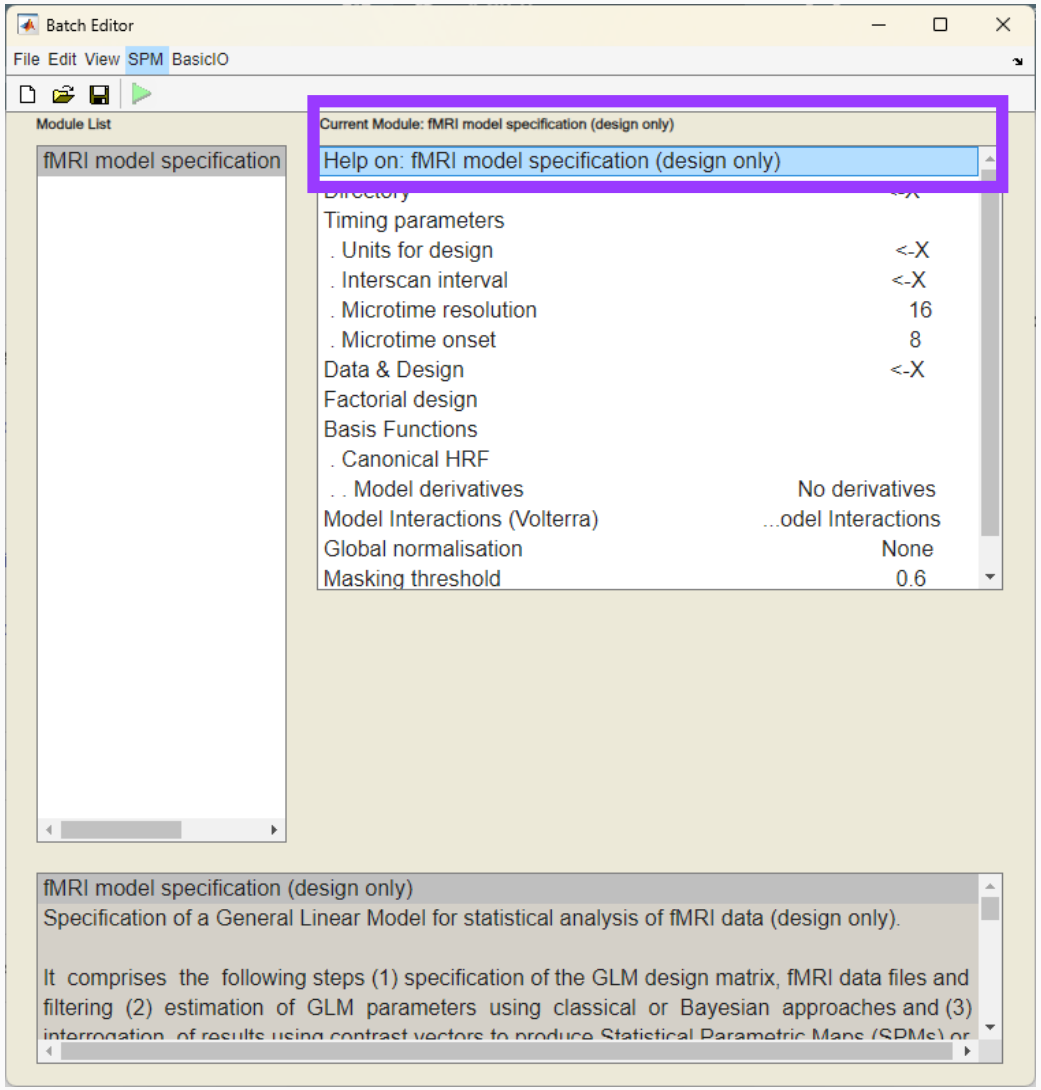
Time domain

Frequency domain

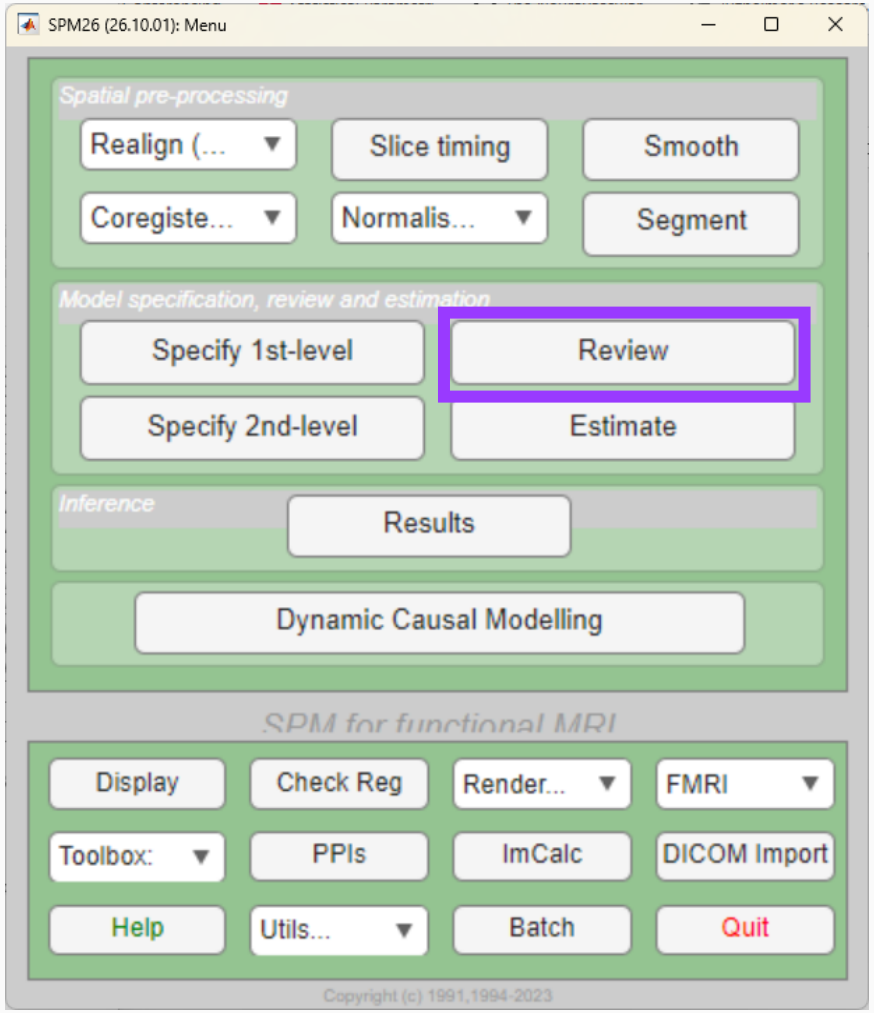


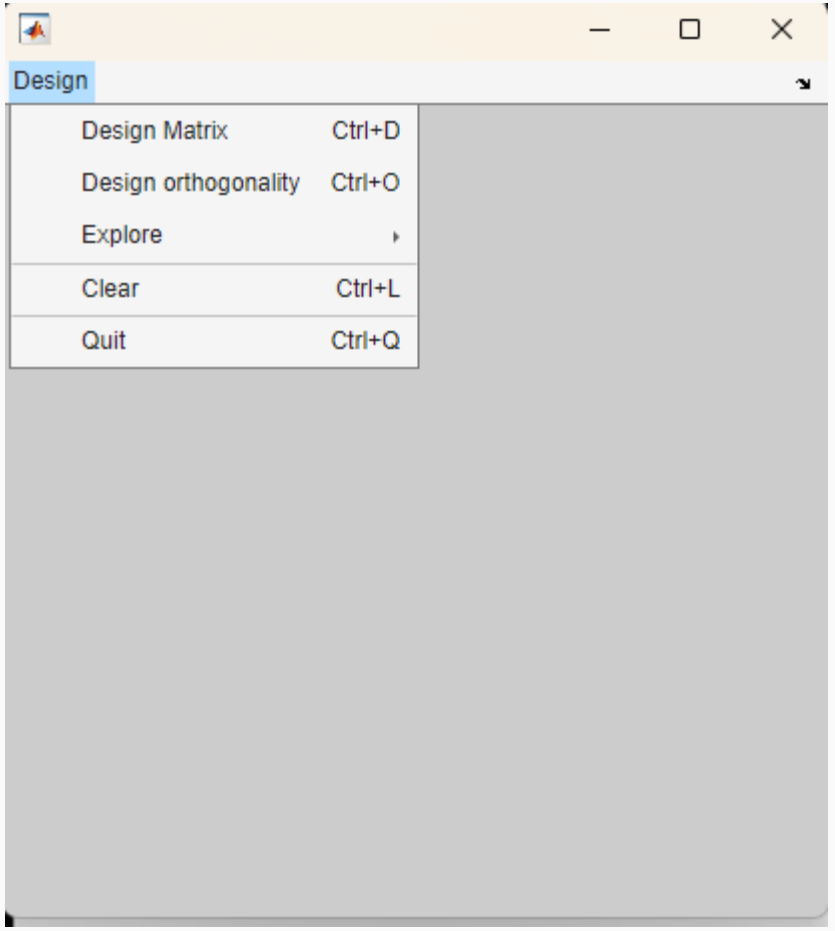
https://imaging.mrc-cbu.cam.ac.uk/imaging/DesignEfficiency

You can specify an SPM before collecting data

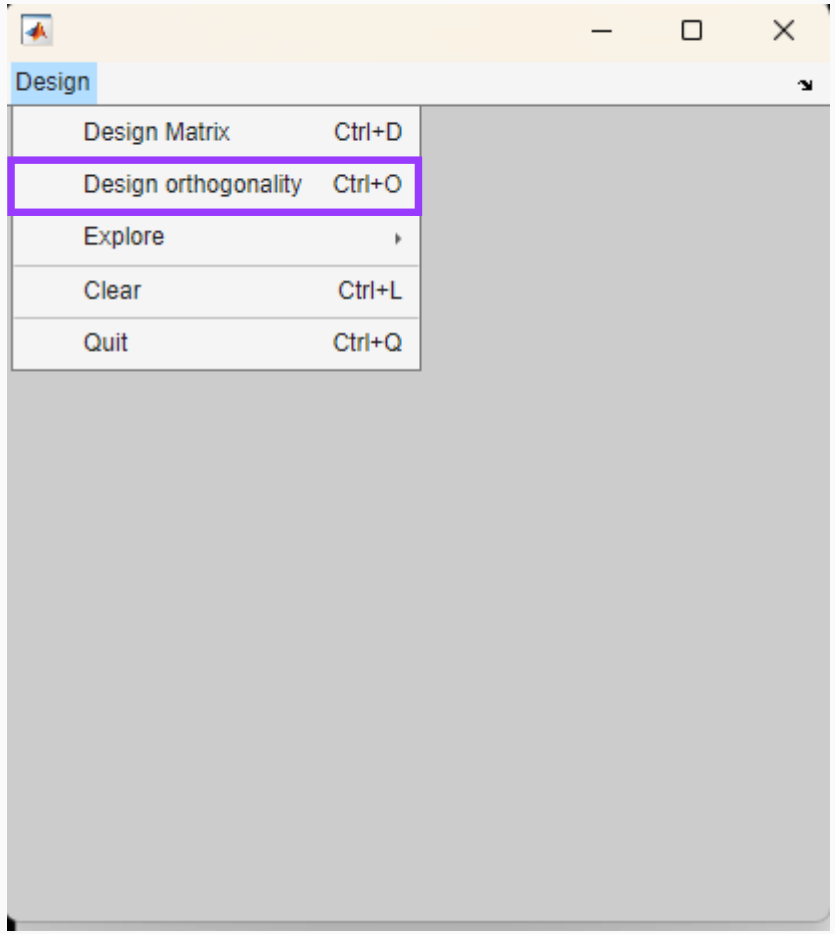


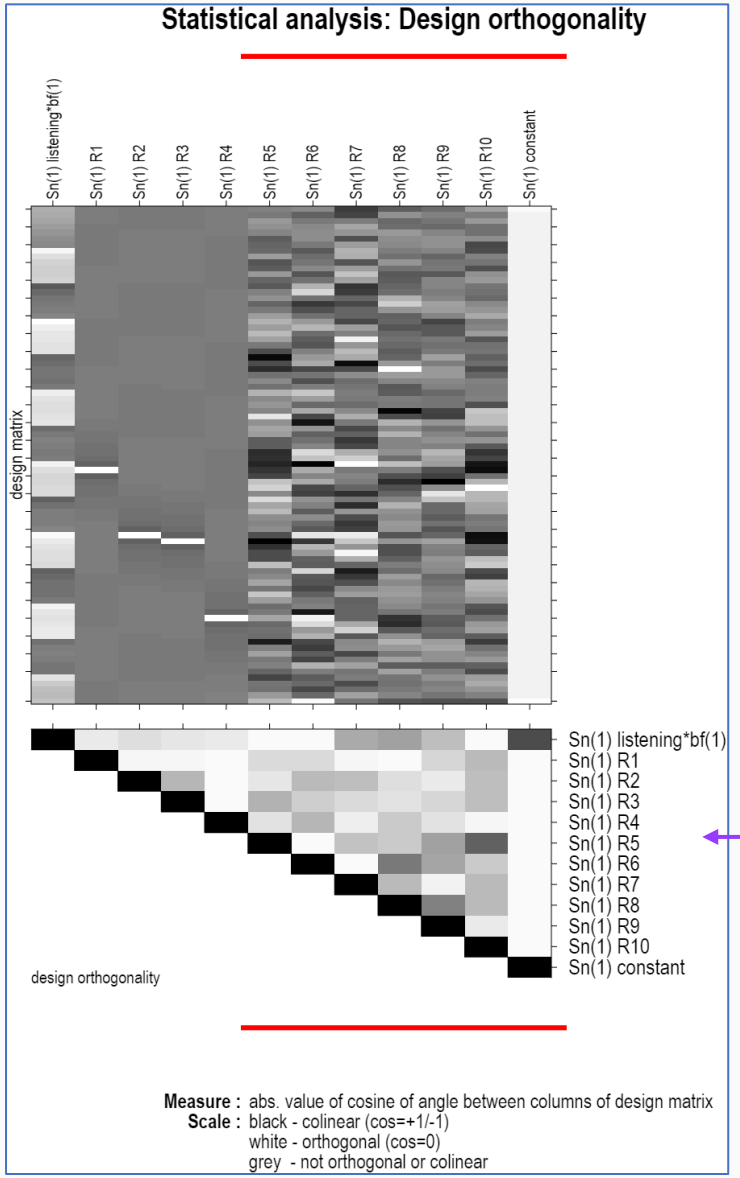
Reviewing the design





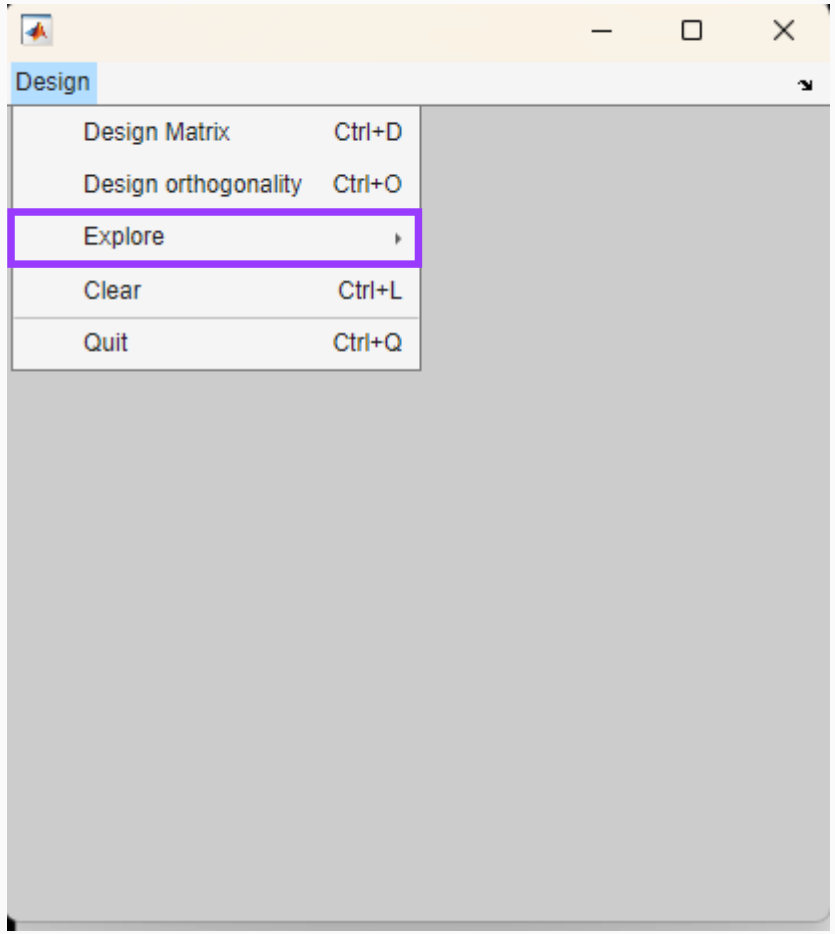
Reviewing the design

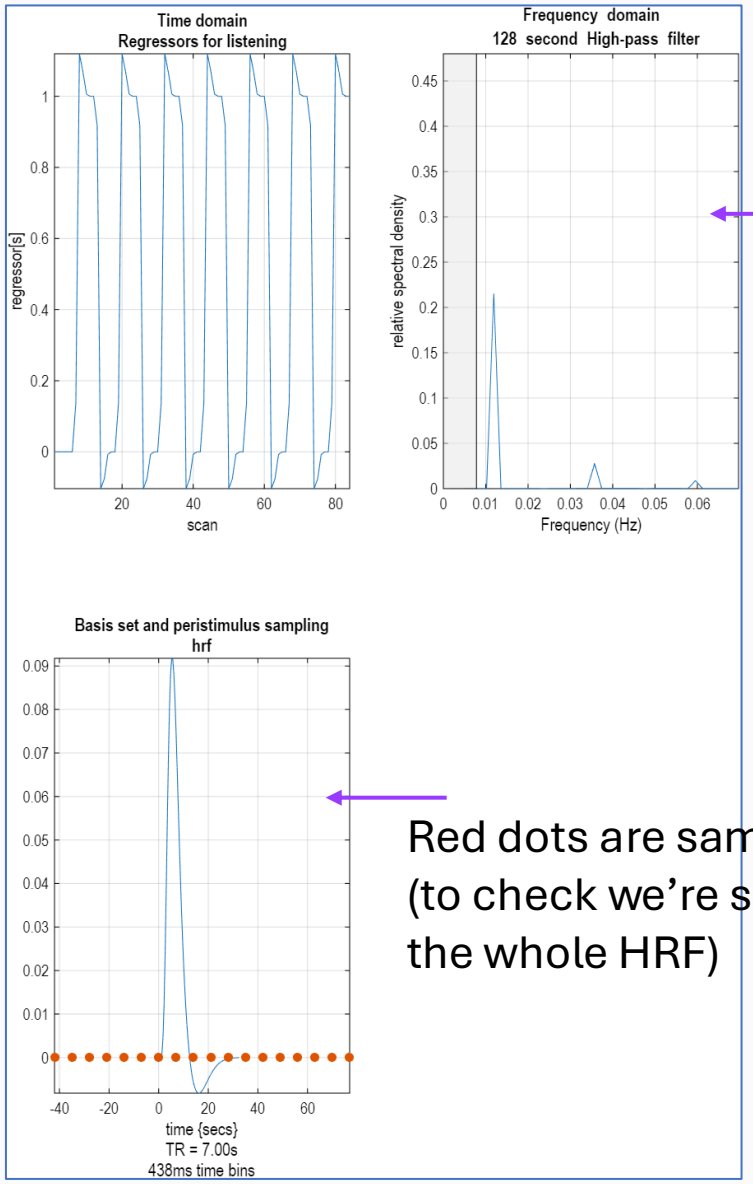




Orthogonality
(Darker is more colinear)

Reviewing the design





Stimulus timing
(to check
not to fast or slow)

Red dots are samples
(to check we're sampling
the whole HRF)

Further reading



Great resource here: link.



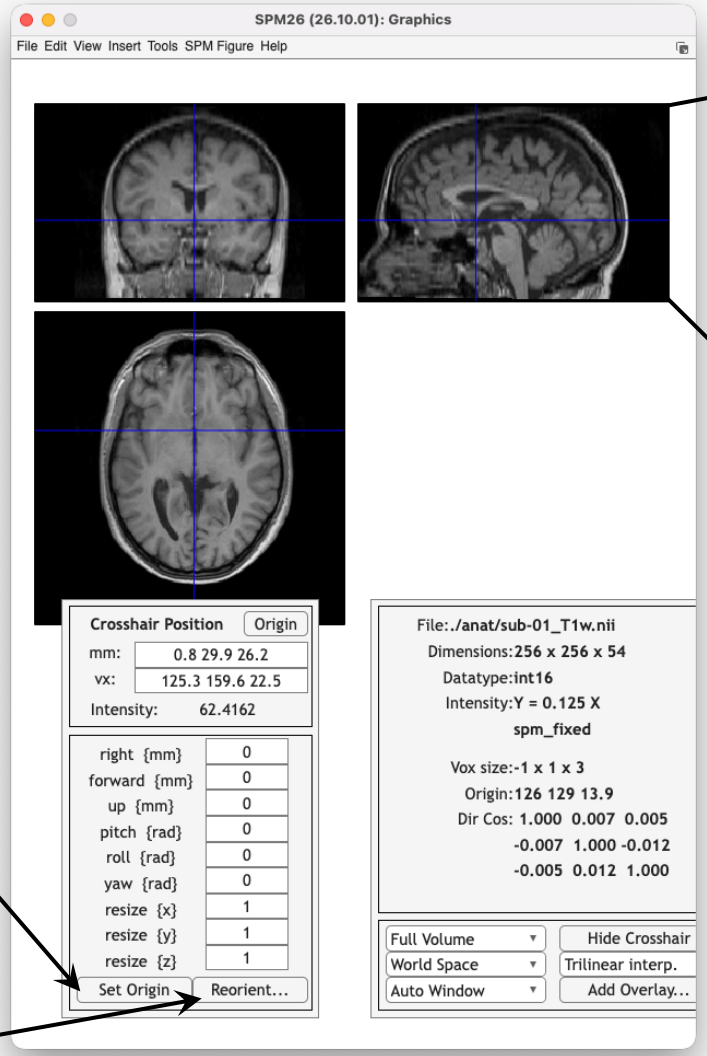
Bonus content

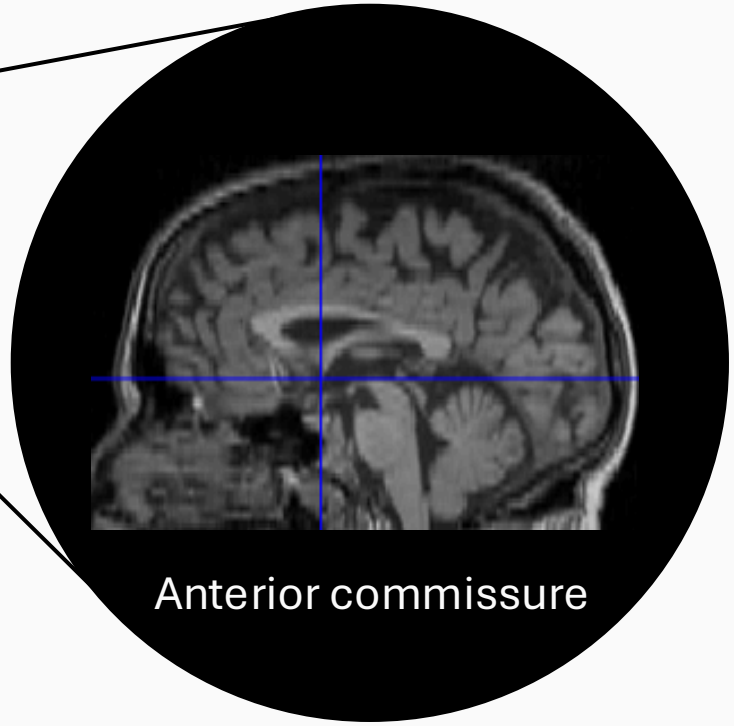
Fixing coregistration errors through image orientation

Manual changes to image orientation

Setting image origin and reorienting images







Change image origin to the crosshair position

Save the transformation

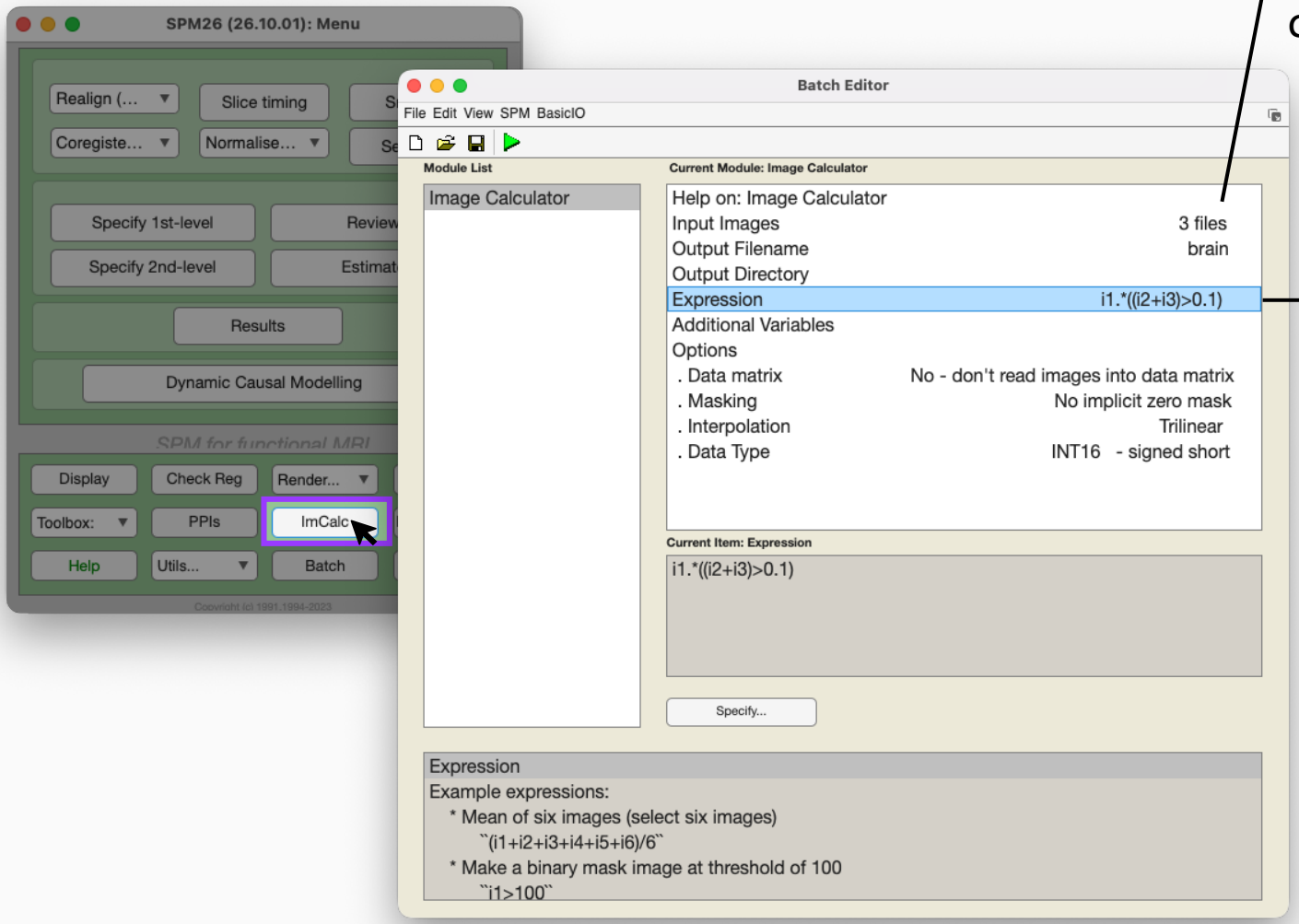


Bonus content

Image calculations with ImCalc

Image manipulation with ImCalc

Skull stripping



`msub-01_T1w.nii` bias corrected structural
`c1sub-01_T1w.nii` segmented grey matter
`c2sub-01_T1w.nii` segmented white matter

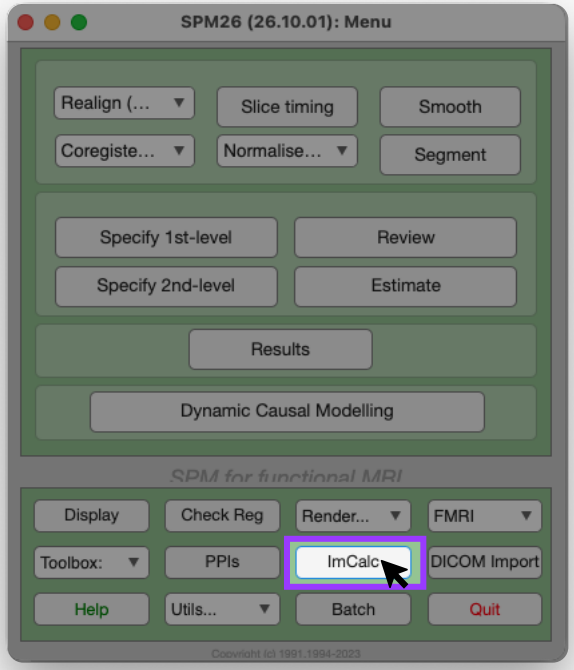
`i1.*((i2+i3)>0.1)`

`i`*n*	*n*th image in the image input order
`.*`	multiply
`>`*x*	threshold resultant image at *x*

Image manipulation with ImCalc

Other useful expressions

`i1>0`	binary mask
`i1>x`	binary mask at *x* threshold
`i1+i2+i3/3`	mean of three images
`i1+i2+i3`	sum of three images
`i1-i2`	difference
`i1.*(i2>0)`	make a binary mask of the second image and apply it to the first



Input images can be read into a matrix, where **X** refers to the matrix:

`mean(X)`	mean of all images
`sum(X)`	sum of all images

